



BSW Insulin Formulary

All registered and unregistered healthcare professionals involved in the prescribing, supply or administration of insulin have a responsibility to be trained and competent in the use of insulin and insulin administration equipment.

It is an NHS HEE requirement that insulin safety training be undertaken every 2 years

[The six steps to insulin safety - DiabetesontheNet](#)

For patients on 'non-formulary' insulins (e.g. from out of area) a switch is not required but refer to specialist services if there are concerns over treatment choices.

Ensure that insulins are prescribed by **brand** name and take into account individual circumstances and preferences when prescribing.

When starting an insulin for which a biosimilar is available, use the product with the lowest acquisition cost. [2022]

When people are already using an insulin for which a lower cost biosimilar is available, discuss the possibility of switching to the biosimilar making a shared decision with the patient. [NICE 2022]

GREEN: Appropriate for initiation in both primary and secondary care

AMBER: Suitable for primary care prescribing following specialist initiation or recommendation

Type 1 Diabetes

Basal-Bolus

NICE recommends a basal-bolus regimen for all adults with type 1 diabetes, with twice-daily insulin detemir as the preferred basal insulin. However, where once-daily dosing is clinically appropriate and supports adherence, insulin glargine may be considered as an alternative basal insulin, provided glycaemic targets are achieved and patient preference is taken into account. Do not offer adults newly diagnosed with type 1 non-basal-bolus insulin regimens. [NICE 2022]

Long Acting Basal Insulin

Insulin detemir is 1st choice basal insulin therapy [NICE 2022] but not used 1st line locally as glargine offers convenience with once daily dosing.

Lantus®
(insulin glargine 100units/mL)

Vial x1 £25.69
Cartridge x5 £34.75
Solostar® pen x5 £34.75

Abasaglar®
(biosimilar insulin glargine 100 units/mL)

Cartridge x5 £35.28
KwikPen® x5 £35.28

Semglee®
(biosimilar insulin glargine 100 units/mL)

Pre-filled pens x5 £29.99

Semglee units are exclusive to Semglee and are not interchangeable with other insulin analogues.

Ultra Long Acting Basal Insulin

Consider ultra long acting insulin if there is a particular concern about nocturnal hypoglycaemia or if people need help from a carer or healthcare professional to administer injections. [NICE 2022]

Tresiba®
(insulin degludec 100units/mL)

Cartridges x5 £46.60

Tresiba®
(insulin degludec 200units/mL)

FlexTouch® 200units/mL x3 £55.92
Only for Type 1 diabetic patients with treatment limiting recurrent hypoglycaemia despite optimal treatment prior to insulin pump therapy, patients with recurrent DKA requiring hospital admission due to delayed administration. Patients receiving insulin from a carer where irregular administration times may contribute to hyper/hypoglycaemia

Please ensure the correct product is selected when prescribing.



Type 1 Diabetes (continued)	
Rapid Acting Insulin	
Offer rapid acting insulin analogues that are injected before meals, rather than rapid acting soluble human or animal insulins, for mealtime insulin replacement for adults with type 1 diabetes. Do not advise routine use of rapid acting insulin analogues after meals for adults with type 1 diabetes. [NICE 2022]	
Trurapi® (biosimilar insulin aspart 100 units/mL)	Vial x1 £11.97 Cartridge x5 £19.82 Solostar® pen x5 £21.42
Apidra® (insulin glulisine 100 units/mL)	Vial x1 £16 Cartridge x5 £28.30 Solostar® pen x5 £28.30
Novorapid® (insulin aspart 100 units/mL)	Vial x1 £14.08 Cartridge x5 £28.31 FlexPen® x5 £30.60
Humalog® (insulin lispro 100 units/mL)	Vial x1 £16.61 Cartridge x5 £28.31 KwikPen® x5 £29.46
High Strength Rapid Acting Insulin	
Consider using a high strength rapid acting insulin where severe insulin resistance and requirement to use more than 20 units of rapid acting insulin per day.	
Humalog® (insulin lispro 200 units/mL)	KwikPen® x5 £58.92
Ultra Rapid Acting Insulin	
Consider if post prandial hyperglycaemia.	
Fiasp® (insulin aspart 100 units/mL)	Vial x1 £14.08 Cartridge x5 £28.31
Lyumjev® (insulin lispro 100 units/mL)	Vial x1 £16.61 Cartridge x5 £28.31 KwikPen® x5 £29.46
Lyumjev® 200units/mL is available but not included on the formulary. Ensure the correct product is selected when prescribing.	
Type 2 Diabetes	
NPH (neutral protamine Hagedorn) Insulin	
Injected once or twice daily according to need. [NICE 2022]	
Humulin® I (isophane insulin 100 units/mL)	Cartridge x5 £19.08 KwikPen® x5 £21.70
Short Acting Insulin	
Consider starting both NPH and short acting insulin (particularly if HbA1c ≥ 75), administered either separately or as a pre-mixed (biphasic) human insulin preparation. [NICE 2022]	
Humulin® S (insulin soluble human)	Cartridge x5 £19.08
Rapid Acting Insulin	
A rapid-acting analogue insulin may be considered for some patients with type 2 diabetes where clinically appropriate, in line with local and national guidance. See preparations under Type 1 guidance.	
Pre-mixed (biphasic) NPH Insulin	
Require administration immediately prior to a meal.	
Humulin® M3 (biphasic isophane insulin 100 units/mL)	Cartridge x5 £19.08 KwikPen® x5 £21.70



Type 2 Diabetes (continued)	
Pre-mixed Analogue Insulin	
Consider using if patient unable to inject insulin immediately before a meal, hypoglycaemia is a problem or blood glucose levels rise markedly after meals. [NICE 2022]	
Novomix® 30 (biphasic insulin aspart 100units/mL)	Cartridge x5 £28.79 Flexpen® x5 £29.89
Humalog® Mix 25 (biphasic insulin lispro 100 units/mL)	Cartridge x5 £29.46 KwikPen® x5 £30.98
Humalog® Mix 50	Cartridge x5 £29.46 KwikPen® x5 £30.98

Long Acting Basal Insulin	
Consider if injection frequency needs reducing or recurrent hypoglycaemia restricts lifestyle or achievement of HbA1c target or the person would otherwise need twice daily NPH injections in combination with oral glucose-lowering drugs. [NICE 2022]	
Lantus® (insulin glargine 100 units/mL)	Vial x1 £25.69 Cartridge x5 £34.75 Solostar® pen x5 £34.75
Abasaglar® (biosimilar insulin glargine 100 units/mL)	Cartridge x5 £35.28 KwikPen® x5 £35.28
Semglee® (biosimilar insulin glargine 100 units/mL)	Pre-filled pens x5 £29.99
Semglee units are exclusive to Semglee and are not interchangeable with other insulin analogues.	

Ultra Long Acting Basal Insulin		
Tresiba® (insulin degludec 100units/mL)	Cartridges x5 £46.60	Recurrent episodes of nocturnal hypoglycaemia or people need help from a carer or healthcare professional to administer injections

High Strength Basal Insulin		
Caution is required with high strength insulins to avoid prescribing, dispensing and administration errors.		
Toujeo® (insulin glargine 300 units/mL)	Solostar® pen x3 £32.14	Recurrent hypoglycaemia and dose <80 units.
Toujeo® (insulin glargine 300 units/mL)	Doublestar® pen x3 £64.27	: Insulin resistance (total dose >1.2mg/kg) or dose >80 units.

The SoloStar® pre-filled pen can provide a dose of 1-80 units per single injection in steps of 1 unit. The Doublestar® pen can provide a dose of 2-160 units in one injection in steps of 2 units.

Toujeo DoubleStar® is one of the few insulin devices in the UK that delivers 2 units of insulin per click. This poses a risk of accidental overdose if the wrong Toujeo product is dispensed or if DoubleStar® is used incorrectly by assuming each click equals 1 unit instead of 2.





Type 2 Diabetes (continued)	
Tresiba® <i>(insulin degludec 200 units/mL)</i>	FlexTouch® 200units/mL x3 £55.92 Insulin resistance (total dose >1.2mg/kg) or dose >80units/day.
Review criteria for high strength insulins: - Improved blood glucose management. - Patient satisfaction	
MHRA alert Tresiba: Nuance Communications, Inc.	
	

References

1. Type 1 diabetes in adults: diagnosis and management NG17. Published 2015. Accessed July 2025. Available from <https://www.nice.org.uk/guidance/ng17>
2. Type 2 diabetes in adults: management NG28. Published December 2015 and last updated July 2025. Accessed Feb 2022. Available from <https://www.nice.org.uk/guidance/ng28>
3. Drug Tariff. Accessed July 2025. Available from <https://nhsbsa.nhs.uk/pharmacies-gp-practices-and-appliance-contractors/drug-tariff>

All costs are correct as per Drug Tariff price November 2025.